

# LAXMANA MANESH JONABOINA

Senior Software Engineer & Architect, Machine Learning

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## PROFESSIONAL SUMMARY

Software Engineer with experience building production-grade ML systems and full-stack applications. Proven ability to architect scalable microservices and RAG pipelines supporting predictive analytics. Expert in Python, distributed systems, and CI/CD automation, with a focus on optimizing system performance (99.9% uptime).

## TECHNICAL SKILLS

**Languages:** Python (Expert), C, Java, C#, JavaScript, SQL, VB.NET

**Architecture & Cloud:** Microsoft Azure, Cloud-Native Architecture, Distributed Systems, Event-Driven Architecture, Microservices, API Design, Object-Oriented Programming (OOP)

**ML & Data:** RAG, MongoDB Vector Search, Hugging Face, PyTorch, Analytics, ML Pipelines, Prompt Engineering

**Infrastructure:** Docker, Kubernetes, GitHub Actions, Azure DevOps, Jenkins, Canary Rollouts

**Databases:** PostgreSQL, MongoDB, MySQL, IBM Db2, Elasticsearch, GraphQL

## PROFESSIONAL EXPERIENCE

**Kansas Dept of Transportation (KDOT)** | Applications Developer | Dec 2024 – Present

- Lead architecture design and technical strategy for modernizing legacy enterprise applications, transitioning systems to highly responsive, cloud-ready frameworks using C#, .NET Core, and Python
- Design and implement scalable, distributed data pipelines and microservices utilizing Docker and Kubernetes, maintaining a highly available infrastructure with 99.9% uptime.
- **Leveraged GenAI tools and prompt engineering** to accelerate the software development lifecycle, utilizing AI-assisted coding to increase code coverage and streamline complex refactoring tasks.
- Define and maintain architecture standards by engineering robust CI/CD pipelines via Azure DevOps and GitHub Actions, reducing release cycles by 40% and ensuring stringent code quality and automation.
- Automated tasks with PowerShell, increasing deployment speed and boosting team productivity by 35%.

**Perficient Corp (Client: Harvest Bank)** | Full Stack Developer | Aug 2024 – Nov 2024

- Developed backend features for healthcare applications using C# & Python, ensuring Engineering Excellence through rigorous system testing.
- Integrated **GenAI capabilities** into production-grade, utilizing secure prompt engineering to enhance end-user efficiency while ensuring the protection of sensitive financial data.

**Accenture** | Custom Software Engineering Analyst | Sep 2021 – Apr 2023

- Designed, developed and maintained critical healthcare applications using **C#, ASP.Net and Angular**.
- Engineered Azure DevOps CI/CD pipelines, reducing release cycles by 40% and deployment errors by 30%.
- Optimized data-intensive **healthcare** solutions, ensuring seamless data exchange by aligning technical specifications with **HL7** standards and clinical priorities to drive digital transformation.
- Designed and deployed microservices-based .NET applications on Azure using Azure DevOps for build and release pipelines, cutting deployment time by 40%.

**LTIMindtree** | Software Engineer | Jul 2020 – Aug 2021

- Implemented Jenkins-based canary rollouts, reducing deployment failures by 25%.
- Automated manual operational tasks using Python scripts, reducing engineering hours by 40%.

## TECHNICAL PROJECTS

**AI Medical Research Assistant (RAG & MongoDB)**

- Designed a **full-stack AI solution - RAG system using MongoDB Atlas Vector Search** to index 4,000+ research articles with sub-second latency.
- Demonstrated leadership in the technology stack by implementing complex retrieval logic and ensuring the frontend/backend integration met high-performance standards.

**Mental Health Sentiment Analysis (Hugging Face)**

- Fine-tuned a Transformer model from Hugging Face for sentiment classification
- Developed a real-time UI using React.js and a Python backend to visualize sentiment trends.

## EDUCATION & CERTIFICATIONS

**MS in Computer Science** | Univ. of Central Missouri | *Relevant Coursework: Machine Learning (A), Adv C#(A)*

**B. Tech in Electronics & Comm. Eng.** | Malla Reddy Engineering College, India

**Certifications:** ML Statistical Foundations (Wolfram), Microsoft Certified: SC-900, AZ-900 & AZ-400